WHEN ARE ALL GROUPS OF ORDER n CYCLIC?

KEITH CONRAD

1. Introduction

For a prime number p, every group of order p is cyclic: each element in the group besides the identity has order p by Lagrange's theorem, so the group has a generator. In fact each nonidentity element of the group is a generator.

There are also composite n for which all groups of order n are cyclic, although the proof is not as simple as choosing an arbitrary nonidentity element and expecting it to be a generator. The first such n is 15: every group of order 15 is cyclic. Here is a proof by Jyrki Lahtonen [6]. If G is a group with order 15 then each element of G has order 1, 3, 5, or 15. By the Sylow theorems, G has a unique subgroup of order 3 and a unique subgroup of order 5, so it has 2 elements of order 3, 4 elements of order 5, and of course 1 element of order 1. That leaves $15 - 2 - 4 - 1 = 8$ elements unaccounted for, so they must all have order 15 and any of them is a generator of G. The same argument shows every group of order 35 is cyclic, and more generally every group of order pq where p and q are distinct primes with $p \not\equiv 1 \bmod q$ and $q \not\equiv 1 \bmod p$ is cyclic: the congruences imply there is one p-Sylow subgroup and one q-Sylow subgroup, making the number of elements of order 1, p, or q equal to $1 + (p-1) + (q-1) = p + q - 1$, so the number of remaining elements is $pq - (p+q-1) = (p-1)(q-1)$, which is positive. Each of these remaining elements must have order pq and thus generates the group.

The general question we want to address is: for which positive integers n is every group of order n cyclic? For each n there is a cyclic group of order n, and a group isomorphic to a cyclic group is cyclic, so a more abstract way of posing our question is: for which n are all groups of order n isomorphic? Whatever way the question is formulated, here is the answer.

Theorem 1.1. *For a positive integer n, all groups of order n are cyclic if and only if n is squarefree and, for each pair of distinct primes p and q dividing n, $q \not\equiv 1 \bmod p$.*

A positive integer n fitting the conclusion of Theorem 1.1 is called a *cyclic number*. It vacuously includes 1 and all primes. In Table 1 are the first five cyclic n with 2, 3, and 4 prime factors. The first 61 cyclic n are online at the OEIS: see https://oeis.org/A003277.

2 primes	3 primes	4 primes
$15 = 3 \cdot 5$	$255 = 3 \cdot 5 \cdot 17$	$5865 = 3 \cdot 5 \cdot 17 \cdot 23$
$33 = 3 \cdot 11$	$345 = 3 \cdot 5 \cdot 23$	$7395 = 3 \cdot 5 \cdot 17 \cdot 29$
$35 = 5 \cdot 7$	$435 = 3 \cdot 5 \cdot 29$	$7735 = 5 \cdot 7 \cdot 13 \cdot 17$
$51 = 3 \cdot 17$	$455 = 5 \cdot 7 \cdot 13$	$8645 = 5 \cdot 7 \cdot 13 \cdot 19$
$65 = 5 \cdot 13$	$561 = 3 \cdot 11 \cdot 17$	$10005 = 3 \cdot 5 \cdot 23 \cdot 29$

Table 1. Cyclic numbers with 2, 3, and 4 prime factors.

From the formula for $\varphi(n)$ in terms of the prime factorization of n, the criterion on n in Theorem 1.1 is equivalent to saying

$$(n, \varphi(n)) = 1,$$

which is a convenient way to generate a long list of cyclic numbers using a computer algebra system that knows the φ-function. Necessarily n is odd when $n \geq 3$, since $\varphi(n)$ is even.

Dickson [2, §6] determined in 1905 those n for which all groups of order n are abelian[1], from which Theorem 1.1 is a consequence. The earliest proof focusing specifically on n for which all groups of order n are cyclic (not just abelian) was given by Szele [8] in 1947.

Proving Theorem 1.1 has two directions:

(1) (necessity) if all groups of order n are cyclic then n is squarefree and $q \not\equiv 1 \bmod p$ for all distinct primes p and q dividing n,
(2) (sufficiency) if n is squarefree and $q \not\equiv 1 \bmod p$ for all distinct primes p and q dividing n then all groups of order n are cyclic.

We will prove necessity in Section 2 and prove sufficiency in three ways in Sections 3, 4, and 5. Other proofs of Theorem 1.1 can be found in the references.

2. Necessity of n being a cyclic number

Assume all groups of order n are cyclic. To prove n is squarefree and $q \not\equiv 1 \bmod p$ for all distinct primes p and q dividing n, we want to show for every other n that there is a noncyclic group of order n. Those other n are either (i) not squarefree or (ii) have a pair of prime factors p and q where $q \equiv 1 \bmod p$ (so $q > p$). In the first case we have $p^2 \mid n$ for some prime p, and in the second case we have $pq \mid n$ where p and q are primes with $q \equiv 1 \bmod p$. The following two examples give us noncyclic groups of order p^2 and pq.

Example 2.1. For each prime p, the group $\mathbf{Z}/(p) \times \mathbf{Z}/(p)$ is not cyclic since it has order p^2 while each element has order 1 or p.

Example 2.2. Let p and q be distinct primes with $p < q$ and $q \equiv 1 \bmod p$. The group

$$\mathrm{Aff}(\mathbf{Z}/(q)) = \left\{ \begin{pmatrix} x & y \\ 0 & 1 \end{pmatrix} : x \in (\mathbf{Z}/(q))^\times, y \in \mathbf{Z}/(q) \right\}$$

has order $(q-1)q$. Since $p \mid (q-1)$, by Cauchy's theorem $(\mathbf{Z}/(q))^\times$ contains a g with order p. The matrices in $\mathrm{Aff}(\mathbf{Z}/(q))$ with upper-left entry a power of g form a group of order pq:

$$\left\{ \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} : a \in \langle g \rangle, b \in \mathbf{Z}/(q) \right\}. \tag{2.1}$$

This group is not cyclic since it is not abelian: $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} g & 0 \\ 0 & 1 \end{smallmatrix}\right)$ are in (2.1) and do not commute, as you can check.

With these examples we can prove that if all groups of order n are cyclic then n is a cyclic number.

Proof. If $p^2 \mid n$ for some prime p then the group $\mathbf{Z}/(p) \times \mathbf{Z}/(p) \times \mathbf{Z}/(n/p^2)$ has order n and is not cyclic since it has the noncyclic subgroup $\mathbf{Z}/(p) \times \mathbf{Z}/(p)$. If n has distinct prime factors p and q such that $q \equiv 1 \bmod p$ then the direct product of (2.1) and $\mathbf{Z}/(n/pq)$ has order n and is not cyclic since it has the nonabelian (hence noncyclic) subgroup (2.1). □

[1]Dickson's theorem is that for $n > 1$, all groups of order n are abelian if and only if the prime factorization of n is $p_1^{e_1} \cdots p_r^{e_r}$ where each e_i is 1 or 2 and for $i = 1, \ldots, r$ we have $p_i^{e_i} \not\equiv 1 \bmod p_j$ for all $j \neq i$.

3. First proof of sufficiency of n being a cyclic number

We will now prove every group having order equal to a cyclic number is a cyclic group using induction on cyclic numbers. The result is obvious for groups of order 1, so assume n is a cyclic number with $n > 1$ and all groups having order equal to a cyclic number less than n are cyclic groups. To prove all groups of order n are cyclic, suppose there is a group G of order n that is *not* cyclic. Another way of describing G is as a minimal counterexample: if there is a non-cyclic group whose order is a cyclic number then there is a *non-cyclic group of least order equal to a cyclic number*, and G is such a group by the inductive hypothesis. We will prove G has various properties until we reach a contradiction.

Since all subgroups and quotient groups of G have order dividing n and every factor of a cyclic number is a cyclic number, by the inductive hypothesis all subgroups and quotient groups of G with order less than n are cyclic groups. We will use this multiple times.

Lemma 3.1. *The group G is not abelian.*

Proof. Since $|G|$ is squarefree, $|G| = p_1p_2\cdots p_r$ for distinct primes p_i. By Cauchy's theorem some $g_i \in G$ has order p_i for $1 \leq i \leq r$. Suppose G is abelian. Then the g_i's pairwise commute and have pairwise relatively prime orders, the order of the product $g_1\cdots g_r$ is the product of the orders of the g_i's, so $g_1\cdots g_r$ has order $p_1\cdots p_r = |G|$. Thus G is cyclic, contradicting the defining condition that G is not cyclic. □

Next we strengthen Lemma 3.1 by showing G is very far from being abelian.

Lemma 3.2. *The group G has a trivial center.*

Proof. Let Z be the center of G, so $Z \lhd G$. If $Z \neq \{e\}$ then $|G/Z|$ is a cyclic number less than $|G|$, so G/Z is cyclic. It is a standard result in group theory that if G/Z is cyclic then G is abelian, so our group G is abelian. That contradicts Lemma 3.1, so $Z = \{e\}$. □

For $x \in G$, its *centralizer* is $Z(x) = \{g \in G : gx = xg\}$. If $x \neq e$ then $Z(x) \neq G$ since the center of G is trivial by Lemma 3.2. The rest of our argument will use centralizers a lot.

Lemma 3.3. *For nontrivial x in G, if $y \in Z(x)$ and $y \neq e$ then $Z(y) = Z(x)$.*

Proof. Since $Z(x) \neq G$, $Z(x)$ is cyclic and thus abelian. Therefore if $y \in Z(x)$, all elements of $Z(x)$ commute with y, which makes $Z(x) \subset Z(y)$. Now $x \in Z(y)$ and $y \neq e$, so by similar reasoning $Z(y) \subset Z(x)$, so $Z(x) = Z(y)$. □

Lemma 3.4. *For nontrivial x and x' in G, if $Z(x) \neq Z(x')$ then $Z(x) \cap Z(x') = \{e\}$.*

Proof. We prove the contrapositive. If $Z(x) \cap Z(x') \neq \{e\}$, let y be a non-identity element of $Z(x) \cap Z(x')$. By Lemma 3.3, $Z(y) = Z(x)$ and $Z(y) = Z(x')$, so $Z(x) = Z(x')$. □

For a subgroup H of a finite group G, the number of subgroups of G that are conjugate to H is $|G|/|\mathrm{N}(H)|$, where $\mathrm{N}(H) = \{g \in G : gHg^{-1} = H\}$ is the normalizer of H.

Lemma 3.5. *If $x \in G$ has prime order then $Z(x) = \mathrm{N}(\langle x\rangle)$ and the number of subgroups of G conjugate to $Z(x)$ is $|G|/|Z(x)|$.*

Proof. Let p be the order of x. Since $Z(x)$ is abelian we have $\langle x\rangle \lhd Z(x)$, so $Z(x) \subset \mathrm{N}(\langle x\rangle)$. To prove $\mathrm{N}(\langle x\rangle) \subset Z(x)$, we adapt the argument from [4, Lemma 1].

Let $g \in \mathrm{N}(\langle x\rangle)$, so

$$gxg^{-1} = x^i, \tag{3.1}$$

where $i \not\equiv 0 \bmod p$. For $k \in \mathbf{Z}^+$ conjugate both sides of (3.1) k times by g to get

$$g^k x g^{-k} = x^{i^k}.$$

In this equation set $k = n = |G|$, so $x = x^{i^n}$ and therefore $i^n \equiv 1 \bmod p$. This implies the order of $i \bmod p$ divides n. Also the order of $i \bmod p$ divides $p-1$, a factor of $\varphi(n)$. Since n is a cyclic number, $(n, \varphi(n)) = 1$. Thus the order of $i \bmod p$ is 1, so $i \equiv 1 \bmod p$ and feeding this back into (3.1) gives us $gxg^{-1} = x$, so $g \in Z(x)$.

The number of subgroups of G conjugate to $Z(x)$ is $|G|/|\mathrm{N}(Z(x))|$. Since $|G|$ is squarefree, $\langle x \rangle$ is a p-Sylow subgroup of G. Therefore $\mathrm{N}(Z(x)) = Z(x)$ because the normalizer of every Sylow subgroup is its own normalizer. Thus $|G|/|\mathrm{N}(Z(x))| = |G|/|Z(x)|$. □

Now we are ready to show the minimal counterexample G leads to a contradiction.

Let p be a prime factor of $|G|$ and x be an element of G with order p (Cauchy's theorem). Then Lemma 3.5 tells us $Z(x)$ has $|G|/|Z(x)|$ conjugate subgroups in G, including itself.

Since $|Z(x)| < |G|$, there is a prime q dividing $|G|/|Z(x)|$ and q does not divide $|Z(x)|$ since $|G|$ is squarefree. Let $y \in G$ have order q (Cauchy again), so $|Z(y)|$ is divisible by q while $|Z(x)|$ is not divisible by q.

We will now look at the union of the subgroups of G conjugate to $Z(x)$ or to $Z(y)$:

$$\bigcup_{g \in G} gZ(x)g^{-1} \cup \bigcup_{h \in G} hZ(y)h^{-1}. \tag{3.2}$$

It will turn out that this subset of G has more than $|G|$ elements, a clear contradiction.

Since $gZ(x)g^{-1} = Z(gxg^{-1})$, Lemma 3.4 tells us that different subgroups conjugate to $Z(x)$ intersect trivially. Similarly, different subgroups conjugate to $Z(y)$ intersect trivially. How does a subgroup conjugate to $Z(x)$ compare to a subgroup conjugate to $Z(y)$? They can't be equal since subgroups of the second kind have order divisible by q and subgroups of the first kind do not, so Lemma 3.4 implies subgroups conjugate to $Z(x)$ and subgroups conjugate to $Z(y)$ intersect trivially.

We can now count the size of (3.2). Using Lemma 3.5 and counting the identity element separately,

$$\begin{aligned}
\left| \bigcup_{g \in G} gZ(x)g^{-1} \cup \bigcup_{h \in G} hZ(y)h^{-1} \right| &= 1 + \frac{|G|}{|Z(x)|}(|Z(x)| - 1) + \frac{|G|}{|Z(y)|}(|Z(y)| - 1) \\
&= 1 + |G| - \frac{|G|}{|Z(x)|} + |G| - \frac{|G|}{|Z(y)|} \\
&\geq 1 + |G| - \frac{|G|}{2} + |G| - \frac{|G|}{2} \\
&= 1 + |G|,
\end{aligned}$$

which is a contradiction and that completes our proof.

4. Second proof of sufficiency of n being a cyclic number

Most proofs I have read that show each group with order equal to a cyclic number is a cyclic group ([1, pp. 9–11], [3], [4], and [5]) involve maximal subgroups, where a *maximal subgroup* of a group is a proper subgroup contained in no other proper subgroup. (The proofs in [7], [8], and [9] are based on ideas other than maximal subgroups, *e.g.*, [7] uses Burnside's normal complement theorem.) In this section we will describe the approach via

maximal subgroups, which is similar in many respects to the argument in Section 3, since the subgroups $Z(x)$ for $x \neq e$ in a minimal counterexample G turn out to be the maximal subgroups of G. We will use Lemmas 3.1 and 3.2 from Section 3, but otherwise develop what we need from scratch.

Here is the strategy. If G is a minimal counterexample then its proper subgroups are all cyclic, so all maximal subgroups of G are cyclic. For a maximal subgroup M of a minimal counterexample G we will show that the size of

$$\bigcup_{g \in G} gMg^{-1}$$

is over half the size of G but is not all of G. Then we'll show there is a maximal subgroup M' not conjugate to M, and the union of its conjugate subgroups also fill up over half of G but not all of G. We'll show the conjugate subgroups of M and M' taken together pairwise intersect trivially, so they have over $|G|$ distinct elements and that is a contradiction.

In a group of prime order its trivial subgroup is maximal, and in a group of non-prime order its trivial subgroup is not maximal since, for each element of prime order (they exist by Cauchy's theorem), the subgroup it generates is a proper subgroup containing the trivial subgroup.

Lemma 4.1. *If x is nontrivial in G then $Z(x)$ is a maximal subgroup of G.*

Proof. Since $x \neq e$ and G has a trivial center (Lemma 3.2), $Z(x)$ is a proper subgroup of G. To prove $Z(x)$ is a maximal subgroup of G, suppose $Z(x) \subset H \subset G$ for a proper subgroup H. Since $|H| < |G|$, the subgroup H is cyclic, and hence abelian, so all of its elements commute with each other. Thus $y \in H \Rightarrow y \in Z(x)$, so $H \subset Z(x)$. Thus $H = Z(x)$. □

Lemma 4.2. *If M is a maximal subgroup of G then $M \neq \{e\}$ and $M = Z(x)$ for each nontrivial x in M.*

This is like Lemma 3.3.

Proof. The subgroup M is nontrivial since $|G|$ is not 1 or prime, and since M is cyclic its elements all commute with each other. So for x in M we have $M \subset Z(x)$. By the definition of maximal subgroups, $M \subset Z(x) \subset G \Rightarrow Z(x) = M$ or $Z(x) = G$. If $Z(x) = G$ then $x \in Z(G)$, and $Z(G)$ is trivial by Lemma 3.2, so $x \neq e \Rightarrow Z(x) = M$. □

Lemma 4.3. *If M and M' are different maximal subgroups of G then $M \cap M'$ is trivial.*

This is like Lemma 3.4.

Proof. We prove the contrapositive. If $M \cap M'$ is not trivial, let x be a non-identity element of $M \cap M'$. By Lemma 4.2, $M = Z(x)$ and $M' = Z(x)$, so $M = M'$. □

Lemma 4.4. *There are no normal subgroups in G other than $\{e\}$ and G.*

This is not like any lemma in Section 3, but it will substitute for the property $\mathrm{N}(P) = P$ of Sylow subgroups that was used in Section 3.

Proof. Let N be a proper normal subgroup of G, so N is cyclic, say of order m. For each $g \in G$ we have $gNg^{-1} = N$, so we can associate to each $g \in G$ the conjugation function $\gamma_g \colon N \to N$ by $\gamma_g(x) = gxg^{-1}$. Each γ_g is an automorphism of N (its inverse is $\gamma_{g^{-1}}$), so $g \mapsto \gamma_g$ is a homomorphism $G \to \mathrm{Aut}(N) \cong (\mathbf{Z}/(m))^\times$.

Let K be the kernel of that homomorphism, so G/K embeds into $\mathrm{Aut}(N)$. Thus $|G/K|$ divides $\varphi(m)$, which divides $\varphi(n)$ since $m \mid n$ (look at the formula for the φ-function, especially on squarefree numbers). Also $|G/K|$ divides $|G|$, which is n. Since n and $\varphi(n)$ are relatively prime and $|G/K|$ divides both, G/K is trivial. Thus $G = K$, which means every element of G conjugates like the identity on the elements of N. Thus $N \subset Z(G)$, so N is trivial by Lemma 3.2. □

Pick $x \neq e$ in G and set $M = Z(x)$, which is a maximal subgroup. Each gMg^{-1} has order $|M|$. Also gMg^{-1} is a maximal subgroup of G, either by checking for all finite groups that the conjugate of a maximal subgroup is a maximal subgroup, or by checking in our special case that $gMg^{-1} = gZ(x)g^{-1} = Z(gxg^{-1})$ and using Lemma 4.1. The number of different subgroups gMg^{-1} as g varies is $|G|/|\mathrm{N}(M)|$, and conjugate subgroups of M intersect trivially when they are distinct by Lemma 4.3, so by counting the identity element separately,

$$\left|\bigcup_{g\in G} gMg^{-1}\right| = 1 + \frac{|G|}{|\mathrm{N}(M)|}(|M| - 1).$$

Since $M \subset \mathrm{N}(M) \subset G$, $\mathrm{N}(M)$ is M or G by maximality of M. From $M \neq \{e\}$ (Lemma 4.2) and $M \neq G$, M is not normal in G (Lemma 4.4), so $\mathrm{N}(M) \neq G$. Thus $\mathrm{N}(M) = M$,[2] so

$$\left|\bigcup_{g\in G} gMg^{-1}\right| = 1 + \frac{|G|}{|M|}(|M| - 1) = 1 + \left(1 - \frac{1}{|M|}\right)|G| \geq 1 + \frac{|G|}{2}.$$

That's a lower bound. We also have an upper bound:

$$\left|\bigcup_{g\in G} gMg^{-1}\right| = 1 + \left(1 - \frac{1}{|M|}\right)|G| < 1 + \left(1 - \frac{1}{|G|}\right)|G| = |G|.$$

By this strict inequality, there is some $x' \in G$ that is not in any conjugate subgroup of M.[3] Set $M' = Z(x')$. Since $x' \neq e$, by reasoning as above with conjugate subgroups of M' in place of M, we get

$$\left|\bigcup_{h\in G} hM'h^{-1}\right| \geq 1 + \frac{|G|}{2}.$$

Subgroups of G having the form gMg^{-1} or $hM'h^{-1}$ are maximal. Such subgroups can't be equal, since otherwise M' is conjugate to M but $x' \in M'$ and x' is (by definition) in no conjugate subgroup of M. Thus every gMg^{-1} and $hM'h^{-1}$ intersect trivially (Lemma 4.3), so by counting the identity element separately,

$$\begin{aligned}\left|\bigcup_{g\in G} gMg^{-1} \cup \bigcup_{h\in G} hM'h^{-1}\right| &= 1 + \left|\bigcup_{g\in G} gMg^{-1}\right| - 1 + \left|\bigcup_{h\in G} hM'h^{-1}\right| - 1 \\ &\geq 1 + \frac{|G|}{2} + \frac{|G|}{2} \\ &= 1 + |G|,\end{aligned}$$

which is a contradiction.

[2]The step analogous to this in Section 3 is that $\mathrm{N}(P) = P$ when P is a Sylow subgroup of G.

[3]In fact, for every finite group G and proper subgroup H, the union of all gHg^{-1} is not G.

5. Third proof of sufficiency of n being a cyclic number

For our next proof that a group whose order is a cyclic number must be a cyclic group, we will use a description of all groups with squarefree order.

Theorem 5.1. *A group has squarefree order if and only if it is a semidirect product of cyclic groups with relatively prime squarefree orders: it is isomorphic to some $\mathbf{Z}/(a) \rtimes \mathbf{Z}/(b)$ where a and b are relatively prime squarefree integers.*

Proof. ($\Longleftarrow$) When a and b are relatively prime and squarefree, every semidirect product of $\mathbf{Z}/(a)$ and $\mathbf{Z}/(b)$ is a group with squarefree order ab.

($\Longrightarrow$) See Theorem 10.2 and the appendix in https://kconrad.math.uconn.edu/blurbs/grouptheory/semidirect-product.pdf. □

Let $|G| = n$ where n is a cyclic number. Since n is squarefree, by Theorem 5.1 we have $G \cong \mathbf{Z}/(a) \rtimes_f \mathbf{Z}/(b)$ where $(a, b) = 1$ and $f : \mathbf{Z}/(b) \to \mathrm{Aut}(\mathbf{Z}/(a)) = (\mathbf{Z}/(a))^\times$ is a homomorphism. Thus $n = ab$ and $\varphi(n) = \varphi(a)\varphi(b)$ due to a and b being relatively prime. Since $\varphi(a) \mid \varphi(n)$, $b \mid n$, and $(n, \varphi(n)) = 1$, we have $(b, \varphi(a)) = 1$. That means $\mathbf{Z}/(b)$ and $(\mathbf{Z}/(a))^\times$ have relatively prime orders, so $f : \mathbf{Z}/(b) \to (\mathbf{Z}/(a))^\times$ is the trivial homomorphism. Thus the semidirect product $\mathbf{Z}/(a) \rtimes_f \mathbf{Z}/(b)$ is a direct product: $G \cong \mathbf{Z}/(a) \times \mathbf{Z}/(b)$. Since $(a, b) = 1$, we have $\mathbf{Z}/(a) \times \mathbf{Z}/(b) \cong \mathbf{Z}/(ab)$, so G is cyclic.

This proof that groups whose order is a cyclic number are cyclic groups looks much shorter than those in the previous two sections, but that impression is deceiving since we did not include a proof of ($\Longrightarrow$) in Theorem 5.1. Including that proof here would make the overall proof in this section just as long as the previous proofs.

References

[1] L. Crew, On the characterization of the numbers n such that any group of order n has a given property P, https://arxiv.org/pdf/1501.03170.pdf.

[2] L. E. Dickson, Definitions of a group and a field by independent postulates, *Trans. Amer. Math. Soc.* **6** (1905), 198–204. URL http://www.ams.org/journals/tran/1905-006-02/S0002-9947-1905-1500706-2/S0002-9947-1905-1500706-2.pdf.

[3] J. Gallian and D. Moulton, When is Z_n the only group of order n?, *Elem. Math.* **48** (1993), 117–119. URL http://gdz.sub.uni-goettingen.de/dms/load/img/?PID=GDZPPN002083140.

[4] Y. Ge, All Groups of Order n are cyclic iff. . ., https://yiminge.wordpress.com/2009/01/22/all-groups-of-order-n-are-cyclic-iff/.

[5] D. Jungnickel, On the Uniqueness of the Cyclic Group of Order n, *Amer. Math. Monthly* **99** (1992), 545–547. URL https://www.jstor.org/stable/2324062.

[6] J. Lahtonen, answer at https://math.stackexchange.com/questions/67407/group-of-order-15-is-abelian.

[7] Y. Sharifi, Groups of order n with gcd(n,phi(n))=1 are cyclic, https://ysharifi.wordpress.com/2010/12/13/groups-of-order-n-with-gcdn-phin1-are-cyclic/.

[8] T. Szele, Über die endlichen Ordnungszahlen, zu denen nur eine Gruppe gehört, *Comm. Math. Helv.* **20** (1947), 265–267. URL https://eudml.org/doc/138922.

[9] S. K. Upadhyay and S. D. Kumar, Existence of a Unique Group of Finite Order, *The Mathematics Student* **81** (2012), 215–218.